Constraining turbulence in the intracluster medium with the radio SZ effect

by

AFIQ ABDUL HAMID

CURRENT DEGREES

Candidate's ORCID

A thesis submitted to the Victoria University of Wellington in partial fulfillment of the requirements for the degree of Doctor of Philosophy

Te Herenga Waka - Victoria University of Wellington

2027

Abstract

Gas motions. SZ effect Pressure fluctuations. Stable methodology. modedling. simulation and some interferometry.

Declaration by author

This thesis is composed of my original work, and contains no material previously published or written by another person except where due reference has been made in the text. I have clearly stated the contribution by others to jointly-authored works that I have included in my thesis.

I have clearly stated the contribution of others to my thesis as a whole, including statistical assistance, survey design, data analysis, significant technical procedures, professional editorial advice, financial support and any other original research work used or reported in my thesis. The content of my thesis is the result of work I have carried out since the commencement of my higher degree by research candidature and does not include a substantial part of work that has been submitted to qualify for the award of any other degree or diploma in any university or other tertiary institution. I have clearly stated which parts of my thesis, if any, have been submitted to qualify for another award.

I acknowledge that an electronic copy of my thesis must be lodged with the University Library and, subject to the policy and procedures of Te Herenga Waka, the thesis be made available for research and study in accordance with the Copyright Act 1968 unless a period of embargo has been approved by the Dean of the Graduate School.

I acknowledge that copyright of all material contained in my thesis resides with the copyright holder(s) of that material. Where appropriate I have obtained copyright permission from the copyright holder to reproduce material in this thesis and have sought permission from co-authors for any jointly authored works included in the thesis.

Publications included in this thesis

Start this section on a new page [this template will automatically handle this].

If you choose to include publications as part of your thesis as described in UQ policy (PPL 4.60.07 Alternative Thesis Format Options) use this section to detail accepted or in press publication/s using the standard citation format for your discipline.

Papers submitted for publication and awaiting review should appear in the next section, **Submitted manuscripts included in this thesis**.

On the page immediately preceding the chapter that includes your publication, in no more than one (1) page, describe your contribution to the authorship if you are not a sole author. In describing your contribution, you must satisfy the University's authorship policy (PPL 4.20.04 Authorship). Authorship is based on having made a substantive contribution to at least one, and usually more than one, of the following activities:

1. conception and design of the project;
2. analysis and interpretation of the research data on which the publication is based;
3. drafting significant parts of the publication or critically reviewing it so as to contribute to the interpretation.

As an author, you must have participated sufficiently in the publication to take public responsibility for at least that part of the work that you contributed.

It may be useful to refer to specific parts of the methods, analyses, results, or discussion to illustrate your contribution to the paper.

If you have not included any of your publications in the thesis then state "No publications included".

Example:

1. [**?**] **Your Name**, Co-author 1, and Final Author, Title of your paper, *Journal*, Issue, Number, Year
2. [**?**] **Your Name**, Co-author 1, and Final Author, Title of your paper, *Journal*, Issue, Number, Year

Submitted manuscripts included in this thesis

List manuscript/s submitted for publication here. As described above for **Publications included in the thesis**, on the page immediately preceding the chapter that includes the submitted manuscript, in no more than one (1) page, detail your contribution to the authorship if you are not the sole author.

If you have no submitted manuscripts from your candidature then state "No manuscripts submitted for publication".

Example:

1. [**?**] **Your Name**, Co-author 1, and Final Author, Title of your paper, submitted to *Journal* on 4th June 2018.

Other publications during candidature

List other publications arising during your candidature using the standard citation format for your discipline. Divide your publications into sub-sections as appropriate in your discipline *e.g.* peer-reviewed papers, book chapters, conference abstracts. Papers submitted for publication and awaiting review are not considered publications and cannot be included in this section.

If you have no publications from your candidature then state "No other publications".

Example:

Conference abstracts

1. [**?**] **Your Name**, Co-author 1, and Final Author, Title of your conference paper, *Proceedings of Conference*, other details.

Book chapters

1. [**?**] **Your Name**, Co-author 1, and Final Author, Title of your chapter, Book, editor, *etc.*.

Contributions by others to the thesis

List the significant and substantial inputs made by others to the research, work and writing represented and/or reported in the thesis. These could include significant contributions to: the conception and design of the project; non-routine technical work; analysis and interpretation of research data; drafting significant parts of the work or critically revising it so as to contribute to the interpretation.

If no one contributed significantly then state “No contributions by others”.

Statement of parts of the thesis submitted to qualify for the award of another degree

The thesis must be comprised only of research undertaken while enrolled in the HDR program unless otherwise approved by the Dean, Graduate School in advance of submission.

If you have been given permission to include your previous work that has been used towards another degree, you must list the relevant parts of the thesis that incorporates this work including, the degree name, year and institution, and the outcome of the submission of material.

If no parts of the thesis have been submitted in this way then state “No works submitted towards another degree have been included in this thesis”.

Research involving human or animal subjects

All research involving human or animal subjects requires prior ethical review and approval by an independent review committee. At UQ, the relevant committee for research involving human subjects is the Human Ethics Unit and the relevant committee for research involving animal subjects is the relevant Animal Ethics Committee. Please provide details of any ethics approvals obtained including the ethics approval number and name of approving committees. A copy of the ethics approval letter must be included in the thesis appendix.

If no human or animal subjects were involved in this research please state: “No animal or human subjects were involved in this research”.

Acknowledgments

Praise to Allah that has given me strength. Appreciation to my parents for their support. Thanks to my supervisors for their patience.

Financial support

Start this section on a new page [the template will handle this for you].

If you are the recipient of an Australian Government Research Training Program (RTP) scholarship, you are required to acknowledge this contribution. Please include the text below:

"This research was supported by an Australian Government Research Training Program Scholarship"

If you received any other financial support for your project, you are also required to acknowledge the funding body/bodies in this section.

If no financial provided then state "No financial support was provided to fund this research".

Keywords

Maximum 10 words; use lower case throughout, separating words/phrases with commas. For example: word, word word, word, word, word word

Australian and New Zealand Standard Research Classifications (ANZSRC)

Provide data that links your thesis to the disciplines and discipline clusters in the Federal Government's Excellence in Research for Australia (ERA) initiative.

Please allocate the thesis a **maximum of 3** Australian and New Zealand Standard Research Classifications (ANZSRC) codes at the **6 digit level** and include the descriptor and a percent weighting for each code. Total percent must add to 100.

> **Example:**
>
> ANZSRC code: 060101, Analytical Biochemistry, 60%
> ANZSRC code: 060104, Cell Metabolism, 20%
> ANZSRC code: 060199, Biochemistry and Cell Biology not elsewhere classified, 20%

Fields of Research (FoR) Classification

Allows for categorisation of the thesis according to the field of research.

Please allocate the thesis a **maximum of 3** Fields of Research (FoR) Codes at the **4 digit level** and include the descriptor and a percent weighting for each code. Total percent must add to 100.

Example:

FoR code: 0601, Biochemistry and Cell Biology, 80%
FoR code: 0699, Other Biological Sciences, 20%

Order for the Remainder of the Thesis

Remainder of the thesis should be in the following order

- Dedications (if applicable)
- Table of Contents
- List of Figures and Tables
- List of Abbreviations used in the thesis
- Main text of the thesis
- Bibliography or List of References
- Appendices

Date of thesis template release: 22 March 2019

For F.N.E. May she live forever.

Contents

Abstract . . . ii

Contents **xi**

List of Figures **xiii**

List of Tables **xiv**

List of Abbreviations and Symbols **xv**

1 Introduction **1**
1.1 Galaxy Clusters . . . 1
1.2 Observables . . . 3
1.2.1 X ray . . . 3
1.2.2 Radio Sunyaev-Zel'dovich effect . . . 4
1.3 Astrophysical Turbulence Scenarios . . . 4
1.4 Turbulence theory . . . 4
1.5 Turbulence inference pipeline . . . 5
1.6 Objectives moving forward . . . 5

2 Assumptions and Transfer Effects **7**
2.1 Smooth SB model . . . 7
2.1.1 Spherical Symmetry . . . 8
2.2 Turbulent Parameters . . . 8
2.3 Deprojection Process . . . 8
2.4 Masking and Apodization . . . 8

3 Abbreviated title **11**
3.1 Non Parametric Methods . . . 11
3.1.1 Iso Contour Surface Brightness . . . 11
3.1.2 Fourier Mode Smooth models . . . 11
3.1.3 Ellipse models . . . 11
3.2 fBm fields and artificial clusters . . . 11

4 Abbreviated title **13**

4.1 Introduction 13

5 Conclusion **15**

A Appendix **17**

A.1 Name of Appendix-1 17

A.2 Name of Appendix-2 17

B Example of Citations **19**

C Example of Code **21**

C.1 Find the greatest number from a list of numbers in *Python* 21

D Example of Equations **23**

E Example of Figures **25**

F Example of Flow Charts **27**

G Example of Tables **29**

List of Figures

E.1 The University Of Queensland 25

F.1 Flow Chart 27

List of Tables

G.1 Name of the Australian Cities 29

List of Abbreviations and Symbols

Abbreviations	
AC	Alternating Current
AFM	Atomic Force Microscopy/Microscope
etc.	*etc.*

Symbols	
$\hat{\rho}$	Density operator
etc.	*etc.*

Chapter 1

Scientific Landscape

> "If you wish to create a Universe from scratch, the final ingredient would be clusters of galaxies."
>
> — [Afiq Hamid 2025]

1.1 Galaxy Clusters

It has been said that mass and time are among the most fundamental parameters in all of astrophysical science. This thesis is principally an exploration of the former through the study of the most massive virialized objects known to exist in the Universe: galaxy clusters.

Galaxy clusters occupy a unique position within the hierarchy of cosmic structures. They represent the endpoint of cosmic structure formation, populating the nodes that connect the filaments of the cosmic web [ref]. The matter composition of galaxy clusters is dominated by a hitherto enigmatic, collisionless dark matter (DM) component ($\sim 80\%$), while the remaining baryonic constituents are made up of the intracluster medium (ICM), a diffuse and turbulent plasma ($\sim 15\%$), with the luminous member galaxies — themselves home to most of the stars, planets, and people within the cluster — make up the small residual ($\sim 5\%$) [ref].

Structurally the dark matter component dictates the overall potential well, setting the stage for all of the baryonic interactions [ref]. At the base of this immense potential well sits the largest member galaxy called the brightest cluster galaxy (BCG), that accounts for 0.5–2% of the cluster baryons. The BCG commonly hosts an active galactic nucleus (AGN), a powerful source of energetic feedback attributed to super massive black hole (SMBH) activity at the origin [ref]. Understanding AGN physics is of broad interest in modern astrophysics, spanning diverse research interests from exotic accretion processes to high powered jet emission [ref], [ref], and while not the primary focus of this work, AGN feedback has some implications for the stirring of turbulence within the ICM.

Permeating the space between the member galaxies in a cluster is the ICM. It is a hot ($T \sim 10^7$–10^8 K), diffuse ($n_e \sim 10^{-3}$–10^{-1} cm^{-3}), optically thin, volume-filling plasma. The ICM is threaded by a weak but dynamically relevant magnetic field ($\sim$ 1–10 μG), detectable via Faraday rotation measures [ref]. As the primary atmosphere of the cluster, the ICM plays an important role in various dynamical processes therein.

- ***AGN thermostat*** The ICM regulates the cooling of gas towards the cluster core — a process that, if left unchecked, would produce excess star formation rates in the BCG. In the absence of a heating mechanism, the dense core would radiate energy efficiently and cool on timescales far shorter than the cluster age, driving a so-called cooling flow [ref]. This cooling is now understood to be moderated by a self-regulating feedback loop: as gas cools and accretes onto the BCG it fuels the central AGN, which injects energy back into the ICM via jets and lobes, suppressing further cooling [ref]. The ICM therefore acts as both the fuel supply and the calorimeter of this baryonic cycle.

- ***Star formation in member galaxies*** The ICM plays an influential role on the evolution and star formation rates of its member galaxies through quenching mechanisms and ram-pressure stripping (RPS). As the satellite galaxies fall into the cluster potential well, contact with the ICM removes internal cold gas reservoirs from the galaxies quenching star formation on relatively shorter timescales especially as the infalling galaxy approaches the denser core ($t \sim 0.3$ Gyr) [ref]. Conversely, on longer timescales ($t \sim 3$ Gyr) strangulation targets the outer hot gas halo and prevents new gas from cooling [ref], [ref].

- ***Cluster mass bias*** Cluster mass measurements can be used as cosmological probes of the large scale matter distribution of the Universe constraining several important parameters of the ΛCDM cosmology model [ref]. Since the DM component cannot be directly observed by any present means, the ICM provides the primary observational avenue of constraining cluster masses via complementary multi wavelength approaches. However, the approaches are primarily dependent on the assumption that the ICM gas density is in approximate hydrostatic equilibrium (HE) within the DM potential well [ref].

The observational leverage we have on all three of the aforementioned processes is mediated through how the ICM manifests across the electromagnetic spectrum. The following section details the primary observational windows through which the ICM is studied in order to derive constraints on the cluster mass, and introduces the key physical quantities and input assumptions that underpin the work done in later chapters.

1.2 Observables

Optical light as the oldest astronomical window was used to initially identify and catalog the first groups and clusters [ref]. It remains a viable avenue to evaluate the matter content of galaxy clusters via gravitational lensing [ref] and velocity dispersion of member galaxies [ref]. However the windows of most relevance towards the mapping of the internal thermodynamics states of the ICM are namely the X-ray and millimeter-wave radio respectively.

1.2.1 X ray

The X-ray window played a vital role in the initial discovery and characterisation of the ICM. To date the ICM of galaxy clusters remain among the brightest X-ray sources on the sky. As unvirialized gas falls into the cluster potential well it is shock heated to temperatures of $T \sim 10^7$–10^8 K, producing emission predominantly via thermal bremsstrahlung (free-free emission), in which electrons are deflected by ions and radiate photons in the process. The X-ray surface brightness S_{X} of the ICM electrons is proportional to the line-of-sight emission integral [ref],

$$S_{\mathrm{X}} \propto \int n_e n_p \Lambda(T)\,\mathrm{d}z \approx \int n_e^2 \Lambda(T)\,\mathrm{d}z, \tag{1.1}$$

where n_e and n_p are the electron and proton number densities respectively, $\Lambda(T)$ is the plasma cooling function, and the integral is taken along the line of sight z. Since the emissivity scales as n_e^2, X-ray observations are particularly sensitive to the dense cluster core, making them well suited for detailed high resolution studies of the inner ICM. However, this sensitivity can be a double-edged sword as the n_e^2 dependence also means that S_{X} can be biased by an unknown gas clumping factor C_ρ arising from inhomogeneous gas distributions in the core [ref].

Since the sound crossing time (represented by the time it takes for pressure fluctuations to travel across a region in the ICM) within an Abell radius of 3 Mpc is only 0.1 - 0.3 of the Hubble time (the timescale of the cluster formation and evolution), it is generally assumed that the ICM will settle into a state of approximate HE:

$$\frac{1}{\rho_g}\frac{dP}{dr} = -\frac{d\phi}{dr} = -\frac{GM(<r)}{r^2}, \tag{1.2}$$

where $M(<r)$ is the total mass enclosed within radius r, ϕ is the gravitational potential, ρ_g is the gas density, and P is Pressure. This local hydrostatic balance contributes to the cluster's broader virialized state and provides a foundational framework through which X-ray observations can be used to infer cluster mass. There has been a good consensus between traditional X-ray methods in finding the true mass with maximum root mean square (RMS) deviation ranging from 5 % to 20 % [ref] [ref]. [However all in not sunshine and roses in the X ray cluster model. Especially when we take a larger picture that clusters are often caught red handed in the act of violent mergers and internal baryon spirals that would make van Gogh look like childs play.].

Another important concept that emerged from the X ray regime is the empirical β-model. By applying an approximation for a self-gravitating collisionless system (originally developed for globular clusters by [ref]) to a radial model of galaxy and gas distribution in equilibrium within the same potential [ref] derived the analytic isothermal radial β model of the ICM.:

$$n_{gas}(r) = n_0 \left[1 + \left(\frac{r}{r_c}\right)^2\right]^{-\frac{3}{2}\beta}, \tag{1.3}$$

where r_c is the radius of the cluster core and $\beta \sim \frac{2}{3}$ is known as the canonical β value [ref]. The profile describes a flat central core transitioning to a power-law decline at large radii. Its broad applicability across the cluster population follows naturally from the self-similar nature of galaxy clusters — that is, clusters are structurally similar objects when scaled by their mass. While modern high-resolution observations and hydrodynamical simulations have revealed limitations of the model — particularly in its assumptions of isothermality and spherical symmetry, and its inability to capture the complexity of merging or dynamically disturbed systems — it remains a robust and widely used first approximation for the ICM radial gas density distribution [ref].

1.2.2 Radio Sunyaev-Zel'dovich effect

The thermal Sunyaev Zel'dovich effect heceforth tSZ is a secondary scattering anisotropy of the cosmic microwave background (CMB). As a result of inverse Compton scattering, Energy levels of incoming photons change via Compton scattering when radio observations appear distorted as either holes or extended sources depending on whether there is a deficit in the number of photons (a decrement of thermodynamic temperature).

$$y = \frac{\sigma}{m_e c^2} \int P dl, \tag{1.4}$$

It is a line of sight projection

Figure 1 shows the bla bla bla. It is a composite image of X ray and tSZ.

1.3 Astrophysical Turbulence Scenarios

Mergers to AGN feedback. All the A. Simonescu stuff. The physical things that create turbulence. Physical scenarios

1.4 Turbulence theory

Energy injection on large scale, scale invariant cascade, and dissipation into heat at small scales. What type of turbulence. This section is all the stuff we have not really delved deep to explore yet.

1.5 Turbulence inference pipeline

This traces all the works done to date. From KG16 to Romero and Pitszi to Saxena.

1.6 Objectives moving forward

Our science Objectives.

The following publication has been incorporated as Chapter **??**.

1. [**?**] **A. Hamid**, Co-author 1, and Final Author, Non parametric model and simulation of SZ surface brightness, *Journal* Issue, Number, Year

If your task breakdown requires further clarification, do so here. Do not exceed a single page.

Chapter 2

Primary Assumptions and Transfer Effects in SZ Surface Brightness Fluctuation Analysis

"Real galaxy clusters exhibit a troubling awareness of our assumptions."

In this chapter I discuss the key assumptions and inputs that go into the typical turbulence inference pipeline discussing their necessity and limitations.

2.1 Smooth SB model

As previously mentioned the first step in order to isolate turbulence from SB data the first step is defining some smooth model. The smooth SB model is a 2D picture of the cluster sans any turbulence. It is essentially a picture of cluster in the state of HE. Consider the Reynolds decomposition for turbulence.

$$S_{\text{obs}}(\theta) = \bar{S}(\theta) + \delta S(\theta)$$

Where $S_{\text{obs}}(\theta)$ is the observed surface brightness, $\bar{S}(\theta)$ is the smooth SB model, and $\delta S(\theta)$ is a small fluctuating component. What you get when you try to find .The challenge is in the method to define

How have smooth models been defined before? Usually some kind of sphere or ellipse tied to a parameter.

What is the problem of these models. We can do elliptical as well. What is problem with that.

2.1.1 Spherical Symmetry

In actuality this phenomena is nothing new. You are not the first to probe it. It has been known to have contributed error a small amount according to Pratt the error budget for this is roughly 1 percent. Haram you write a paper and thesis about cluster without citing the Buote papers. Read those papers, rabbit hole a bit, and cite here.

Galaxy clusters are not spherical, but almost everyone treats them as if they are, because it's computationally far simpler. The question is — how much does this matter, and for what observables?

This is the shit that you gotta curve fit to: This is the zeroth order model. Extension to the zeroth order model is spherical and elliptical averaging.

$$S_X(b) = S_0 \left[1 + \left(\frac{b}{r_c} \right)^2 \right]^{-3\beta+0.5} \tag{2.1}$$

But how we do this is the

2.2 Turbulent Parameters

So once we have the raw fluctuations. we would then take the power spectrum. Then we deproject the power spectrum. We don't really know the parameters. Suffice to say that shit is complex. You have multi scale, multi layer, multi mach numbers. How does one even know what the injection is? Here is some theory at least one page to prove it.

2.3 Deprojection Process

Everything we see is a two dimensional deprojection of a three dimensional phenomena. There is an Abel transfrom used for

2.4 Masking and Apodization

Commonly is a Gaussian window.

Hello I am here. The following publication has been incorporated as Chapter 3.

1. [**?**] **Your Name**, Co-author 1, and Final Author, Title of your paper, *Journal* Issue, Number, Year

If your task breakdown requires further clarification, do so here. Do not exceed a single page.

Chapter 3

Methodologies for Testing Turbulence Recover ability

In this chapter we talk about how we interrogated all the assumptions that go into the pipeline.

3.1 Non Parametric Methods

Without any prior assumptions on turbulent mach numbers.

3.1.1 Iso Contour Surface Brightness

Its a type of harmonics based algorithm

3.1.2 Fourier Mode Smooth models

This is actually a Low Pass Filtering

3.1.3 Ellipse models

This is actually two contributions in one. One is a sniper genetic algorithm method based method. And the other is a drone swarm.

Genetic Algorithms with Differential Evolution

Bayesian inference with emcee

Probably some corner plots here.

3.2 fBm fields and artificial clusters

Add your text here.

Hello I am here. The following publication has been incorporated as Chapter 4.

1. [**?**] **Your Name**, Co-author 1, and Final Author, Title of your paper, *Journal* Issue, Number, Year

If your task breakdown requires further clarification, do so here. Do not exceed a single page.

Chapter 4

Methodologies for Testing Turbulence Recover ability

In this chapter we talk about the assumptions.

4.1 Introduction

Add your text here.

Chapter 5

Conclusion

Conclude your thesis.

Appendix A

Appendix

Write your appendix here. Following two are examples.

A.1 Name of Appendix-1

A.2 Name of Appendix-2

Appendix B

Example of Citations

This text is only for Bibliography testing purposes. This is a book by Cawvey et al. [**?**]. There are few journal articles [**?**, **?**] in this bibliography. This is a proceeding paper [**?**]. This is a PhD thesis by Peerling [**?**]. These are few other types of citations [**?**].

Appendix C

Example of Code

C.1 Find the greatest number from a list of numbers in *Python*

```
a=[1,2,3,4,6,7,99,88,999]
    max= 0
    for i in a:
        if i > max:
            max=i
    print(max)
```

Appendix D

Example of Equations

$$E = mc^2 \tag{D.1}$$

$$a = b + c \tag{D.2}$$

$$x = y + z \tag{D.3}$$

$$\begin{aligned} a = b &+ c \\ &+ d + e \end{aligned}$$

$$sinx + cosx = 1 \tag{D.4}$$

Appendix E

Example of Figures

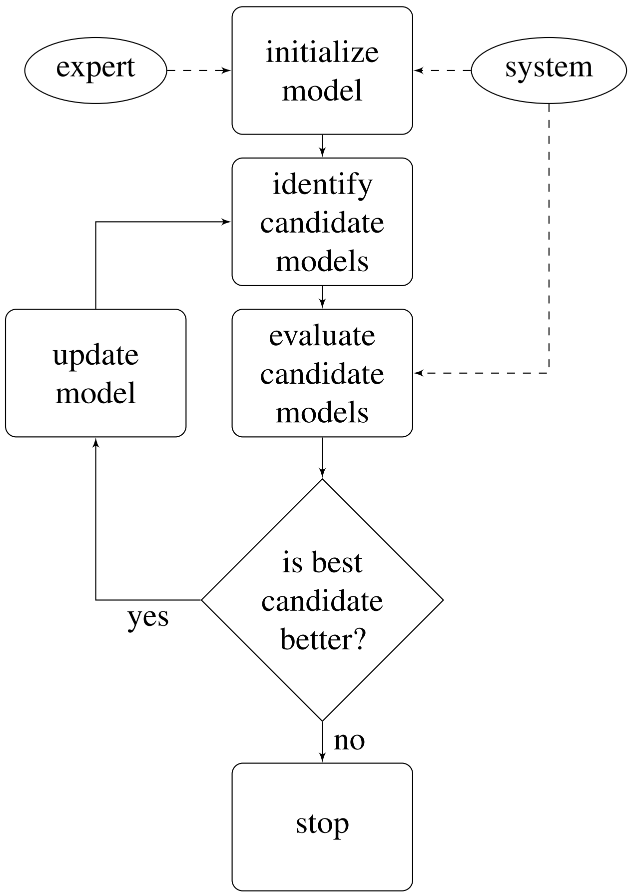
Figure E.1: The University Of Queensland

The Figure E.1 represents beauty of the UQ campus.

Appendix F

Example of Flow Charts

Figure F.1: Flow Chart



Flow chart F.1 is a simple example.

Appendix G

Example of Tables

Here is a really simple table G.1.

Table G.1: Name of the Australian Cities

Number	**Name**
1	Brisbane
2	Sydney
3	Melbourne
4	Canberra
5	Perth
6	Adelaide
7	Hobart
8	Darwin

Endquote goes here.

Author of quote,
Source of quote